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International AS

Physics (9630)
PH02 Electricity, Waves and Particles
Report on the Examination

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REPORT ON THE EXAMINATION: INTERNATIONAL AS PHYSICS (9630) PH02 – JUNE 2025

This paper was found to be slightly more accessible than in previous years, with a moderate increase in the mean; it differentiated well between students, with a larger standard deviation than in some previous series.

As in most previous series, students performed better on average in the multiple-choice section than in Sections A and B. In Sections A and B, students performed best in the one-step and two-step calculation questions (such as 02, 04.2, 09.2, 11.1). Students also did well in taking relevant data from a graph as in 06.1 and 06.2.

As in previous series, students performed less well in descriptions and explanations, including recall questions such as 01.1 and 05.1.

Another area of weakness, as in previous series, was commenting on experimental procedure, such as in 11.4 and 08.2.

08.2 was a 6-mark written question and it differentiated well between students.

Students were also challenged by the final parts of some questions, which required both scientific knowledge and its application to an unfamiliar context, such as 05.2, 08.3, 11.4. These items were designed to be challenging as the final parts of those questions.

Some general principles to note that apply in every series:

- A question beginning ‘Determine’ usually requires some working that demonstrates how the students arrived at the answer. A bald correct answer with no working or incomplete working may not gain all the marks. Two examples in this paper are 06.2 and 09.2.
- When a gradient is determined from a graph, the points chosen must be clearly indicated on the graph and must be further apart than half the range of the data. The example in this paper is 06.2.
- If an explanation-type question is about a physical quantity, it is often necessary to define the quantity as part of the explanation. An example in this paper is 05.1.

QUESTION 01

01.1 was not very successfully answered, with less than half the students scoring the mark. Many students mentioned frequency but not electromagnetic radiation.

01.2 was well answered as a straightforward calculation, with almost two-thirds of the cohort scoring full marks.

QUESTION 02

This was also well answered, with almost two thirds of students scoring full marks in this calculation. Some students did not take the reciprocal of 280 (N) to get the slit spacing, or even confused n with N.

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QUESTION 03

03.1 was fairly well answered, with over half of students scoring the mark. Most students showed some understanding of the context and either referred to the wave equation or stated that the frequency stayed the same, but both were required for the mark.

Due to an unfortunate error in the question paper for 03.2, this question part was discounted from the paper. All students were awarded one mark. The corrected question will appear in the paper due to be uploaded for teachers' and students' access in the future.

QUESTION 04

In 04.1, most students scored one of the two marks. The best students referred to the context: the child moves back and reaches maximum height, then moves forward and reaches maximum height, and this is half a period.

04.2 was very well answered with most students scoring full marks; ecf from 04.1 was given.

04.3 yielded a good range of responses, with all the different possible correct answers seen. We did not give credit for 'the child's centre of mass moves' as of course a simple pendulum's mass also moves back and forth.

In 04.4 most students scored at least one mark but very few scored full marks. Lots of students showed the misconception that the KE graph looks like the modulus of sine rather than the square of sine. We required that the sketch of the sine curve should not have sharp turning points.

QUESTION 05

In 05.1 most students scored only one of the two marks: many students did not define polarisation and only described the difference between longitudinal and transverse waves.

05.2 yielded some good responses and most students gave some reasonable attempts for such a novel question. A large majority of students expressed the relevant idea that mobile phones are held in different orientations. Many also were able to describe the idea of aligning receivers with polarised signals. However, only around 1 in 6 students gave both ideas to get full marks.

QUESTION 06

06.1 was very well answered with almost 9 in 10 students scoring the mark. Some read the graph scale incorrectly.

In 06.2, three-quarters scored at least two marks, finding the gradient correctly and quoting the internal resistance; for the third mark, we required some working to show how the internal resistance is related to the gradient, and fewer students scored this final mark. As mentioned above, a 'Determine' question may require working or derivation.

06.3 was well answered and most students scored at least one mark. Less good responses mixed up the emf and the voltage across the lamp.

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QUESTION 07

Students found Question 7, about refraction and total internal reflection, generally challenging.

Around half of the students scored the mark in 07.1. Some students drew the ray changing direction before leaving the water (at the position of the spout); others thought the ray would refract out of the water.

Again around half of the students scored the mark in 07.2. Some incorrect responses showed the ray refracting upwards out of the water (i.e. towards the normal); some had the ray totally internally reflecting before reflecting off the table.

Some good understanding was shown in 07.3, with almost half of the students scoring at least two of the three marks. Most students understood that the ray no longer totally internally reflected, and many correctly explained about the angle of incidence becoming less than the critical angle. Very few (around 1 in 20) explained why the light refracted in the direction that it did.

07.4 was not very successfully answered, with fewer than half of students scoring the mark. Many students wrote ‘do not look (directly) at the laser (beam)’ but it would be impossible to view a classroom demonstration with this rule. Instead, ‘do not point lasers at people/at people’s eyes’ is a better measure. We did not allow safety glasses, or goggles, unless it was clear that specific laser protection goggles were described.

QUESTION 08

Students found question 8 challenging overall but we did see a good variety of responses and the question differentiated well between students.

Nearly half scored full marks in 08.1, and it was evident that students were prepared for this question.

08.2, the long answer question describing an experiment, yielded a good spread of marks. Around half scored at least two marks. Many students were able to manipulate the $y = mx + c$ equation successfully, and list at least some of the variables and their measuring instruments. Very few successfully identified techniques for reducing uncertainty in the experiment.

08.3 identified the best students, with only 1 in 20 scoring the mark. A common misconception was that there would be less uncertainty in the frequency when more antinodes are seen.

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QUESTION 09

Students found question 9 relatively less challenging.

09.1 was not very successfully answered for what is arguably a ‘standard’ question. Most students scored one mark but less than a third scored two or three marks. A lot of students did not refer to photons at all; some reference to the photon model is necessary in a discussion about line spectra.

09.2 was much better answered, with the majority scoring at least three of the four marks. As mentioned above, we required some working for full marks as this is a ‘Determine’ question.

Over half of students scored the mark in 09.3.

QUESTION 10

Question 10, the data analysis question, was fairly well answered, continuing the trend of improving performance in this section.

Over half of students scored the mark in 10.1. The most common error was suggesting the use of Vernier callipers.

10.2 was not successfully answered, with less than 1 in 10 students scoring the mark. A common error was the misuse of the term ‘calibrate’: calibration is of course verifying the increments across the range of a measuring device rather than just identifying and correcting a zero error, and this would be an unreasonable step to take before using a micrometer. The command word used was ‘describe’ rather than ‘state’ so a brief answer such as ‘ensure there is no zero error’ did not gain the mark as it is not a description of the procedure required.

10.3 was slightly better answered with almost half of students scoring the mark. A common error was simply stating that random errors would be reduced without referring to the context.

10.4 was fairly well answered with two-thirds of students scoring at least two marks. Some students did not identify and remove the anomaly. There was also a significant figure penalty: one or two significant figures is conventional for uncertainties.

10.5 was very well answered with over two thirds of students scoring both marks. Ecf was given, and extra significant figures were condoned here.

QUESTION 11

Question 11 ramped up in difficulty.

Students found 11.1 very accessible with over half scoring all three marks. Some students read the data from the graph incorrectly, but most used the potential divider equation successfully.

11.2 was also fairly well answered, with around half scoring at least two of the three marks. We credited a variety of approaches here but the best answers discussed maintaining the voltage ratio between the two resistances.

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11.3 was found to be slightly harder with just under half scoring the mark. We required some (brief) explanation of the benefit to the user in being able to adjust the resistance.

11.4 was intentionally difficult as the last question part in the section and fewer than 20% of students scored the mark.

SECTION C

The most successfully answered questions were:

12: a simple resistance calculation

20: a Young Modulus calculation

24: a question about fluorescent tubes

The less successfully answered questions were:

13: using data from an I–V characteristic graph. B was a popular distractor.

18: a question about progressive waves. B was a popular distractor.

21: a photoelectric effect question. Both C and D were popular distractors.

23: a modal dispersion question. Both A and B were popular distractors.

There were no questions where a distractor was more popular than the correct answer.

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